

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$7x - 4y = -84$

For the given equation, which table gives three values of x and their corresponding values of y ?

Answer

A.

x	0	4	8
y	21	28	35

B.

x	0	4	8
y	35	28	21

C.

x	21	28	35
y	0	4	8

D.

x	21	28	35
y	8	4	0

Correct Answer: A

Rationale

Choice A is correct. To verify which table represents this linear relationship, the values in each table can be checked by substituting them into the given equation. The table in choice A shows that when $x = 0, y = 21$. Substituting these values into the given equation yields $7(0) - 4(21) = -84$, or $-84 = -84$, which is true. Additionally, the table in choice A shows that when $x = 4, y = 28$. Substituting these values into the given equation yields $7(4) - 4(28) = -84$, or $-84 = -84$, which is true. Finally, the table in choice A shows that when $x = 8, y = 35$. Substituting these values into the given equation yields $7(8) - 4(35) = -84$, or $-84 = -84$, which is true. Therefore, the table in choice A gives three values of x and their corresponding values of y .

Choice B is incorrect. The table in choice B shows that when $x = 0, y = 35$. Substituting these values into the given equation yields $7(0) - 4(35) = -84$, or $-140 = -84$, which is not true.

Choice C is incorrect. The table in choice C shows that when $x = 21, y = 0$. Substituting these values into the given equation yields $7(21) - 4(0) = -84$, or $147 = -84$, which is not true.

Choice D is incorrect. The table in choice D shows that when $x = 21, y = 8$. Substituting these values into the given equation yields $7(21) - 4(8) = -84$, or $115 = -84$, which is not true.